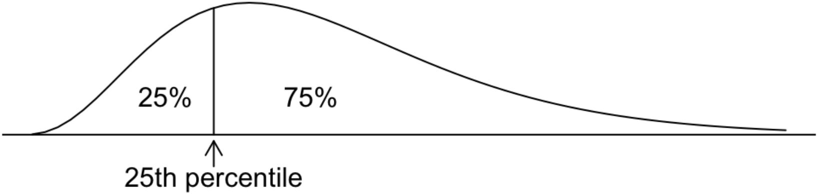
STAT 20000 Chapter 4, Part 2
The Average and Standard Deviation

Five-Number Summary & Boxplots

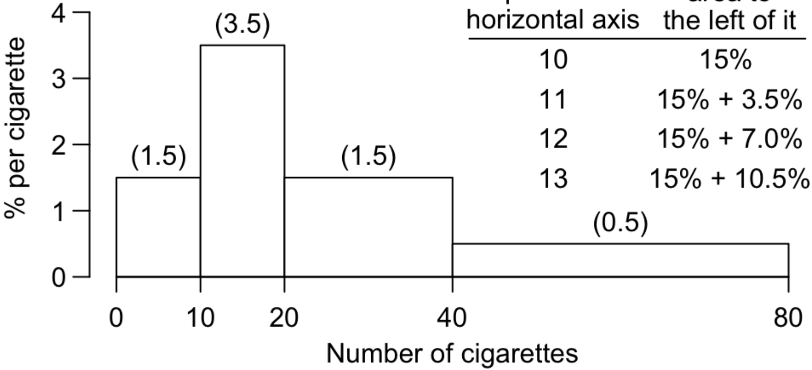
Histograms and Percentiles

What is the 25^{th} percentile of a histogram?



The point on the horizontal axis such that **25%** of the area under the histogram lies to the left of that point (and **75%** to the right).

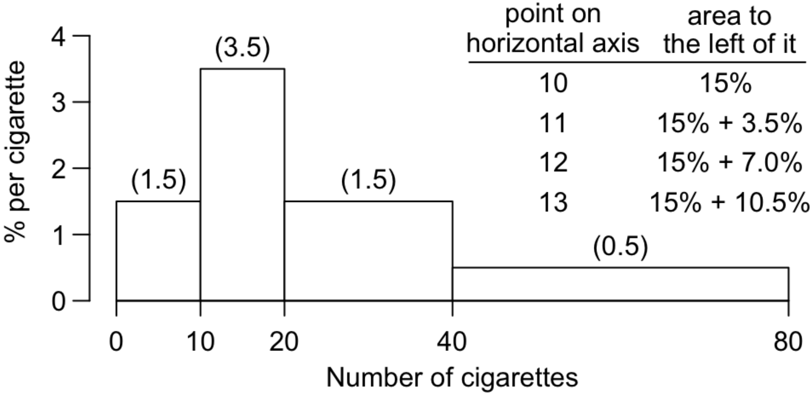
What is the 25^{th} percentile for the cigarette histogram?
(The class intervals include the right endpoint, but not the left.)



point on horizontal axis	area to the left of it
10	15%
11	15% + 3.5%
12	15% + 7.0%
13	15% + 10.5%

So the 25^{th} percentile is about ______.

What is the 25^{th} percentile for the cigarette histogram?
(The class intervals include the right endpoint, but not the left.)



So the 25^{th} percentile is about <u>13</u>.

What is the 25^{th} percentile for the cigarette histogram?
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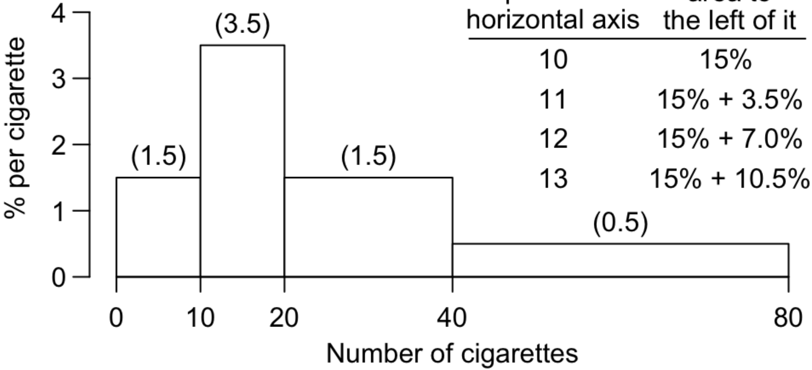

point on horizontal axis	area to the left of it
10	15%
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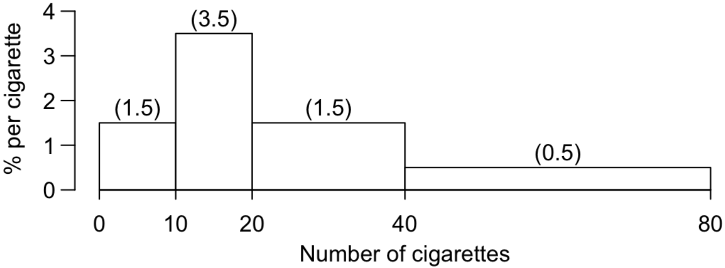
13	15% + 10.5%

So the 25^{th} percentile is about 13.

1 out of 4 of them smoked 13 or fewer cigarettes per day.

Other percentiles are defined in a similar way.
E.g., the 95^{th} percentile is the point on the horizontal axis such that 95% of the area under the histogram lies to the left of it.

Q: What is the 50^{th} percentile for the cigarette histogram?



- 50^{th} percentile = ______.
- _____ out of _____ of these men smoked ______ or fewer cigarettes per day.

Other percentiles are defined in a similar way.
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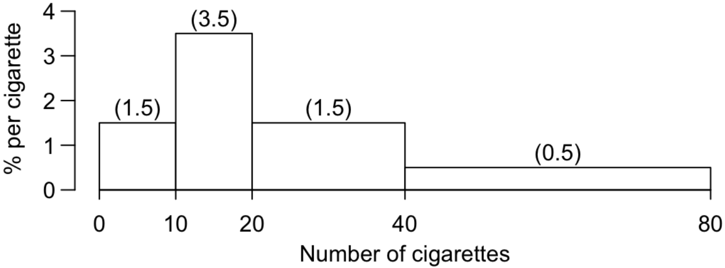

- 50^{th} percentile = ___20___.
- _____ out of _____ of these men smoked ______ or fewer cigarettes per day.

Other percentiles are defined in a similar way.
E.g., the 95^{th} percentile is the point on the horizontal axis such that 95% of the area under the histogram lies to the left of it.

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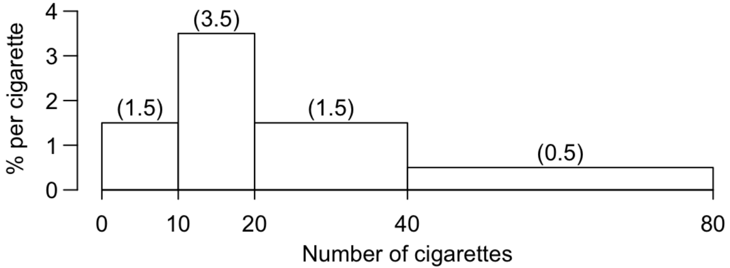
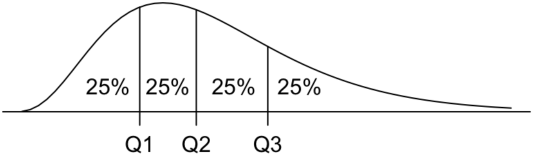


- 50^{th} percentile = __20__.
- __1__ out of __2__ of these men smoked __20__ or fewer cigarettes per day.

Quartiles, IQR, Five-Number Summary

- **Quartiles** divide data into 4 even parts
  - **first quartile $\mathbf{Q_1}$** $=$ 25th percentile:
    25% of data fall below it and 75% above it
  - **second quartile $\mathbf{Q_2}$** $=$ median $=$ 50th percentile
  - **third quartile $\mathbf{Q_3}$** $=$ 75th percentile
    75% of data fall below it and 25% above it



- **Interquartile Range** $(IQR) = Q_3 - Q_1$
- **Five-Number Summary**:

$$\text{min}, \quad Q_1, \quad \text{Median}, \quad Q_3, \quad \text{max}$$

Example 1

For the 9 numbers: 43, 35, 43, 33, 38, 53, 64, 27, 34

$$\begin{array}{c} 43 \\ 35 \\ 43 \\ 33 \\ 38 \\ 53 \\ 64 \\ 27 \\ 34 \end{array} \xrightarrow{\text{sort}} \begin{array}{c} 27 \\ 33 \\ 34 \\ 35 \\ 38 \\ 43 \\ 43 \\ 53 \\ 64 \end{array}$$

- $\mathrm{IQR} = Q_3 - Q_1 = 48 - 33.5 = 14.5$
- Five number summary: 27, 33.5, 38, 48, 64

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For the 9 numbers: 43, 35, 43, 33, 38, 53, 64, 27, 34

$$
\begin{array}{ccccc}
43 & & 27 & & \\
35 & & 33 & & \\
43 & & 34 & & \\
33 & & 35 & & \\
38 & \xrightarrow{\text{sort}} & 38 & \leftarrow & \begin{array}{c}\text{overall}\\ \text{median}\end{array} = Q_2 \\
53 & & 43 & & \\
64 & & 43 & & \\
27 & & 53 & & \\
34 & & 64 & &
\end{array}
$$

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$$
\begin{array}{ccclcl}
43 & & 27 & \left.\begin{array}{l} \\ \\ \\ \\ \end{array}\right\} & \leftarrow \begin{array}{c}\text{median}\\ \text{of this half}\end{array} & = \dfrac{33+34}{2} = 33.5 = Q_1 \\
35 & & 33 & & & \\
43 & & 34 & & & \\
33 & & 35 & & & \\
38 & \xrightarrow{\text{sort}} & 38 & \leftarrow & \begin{array}{c}\text{overall}\\ \text{median}\end{array} & = Q_2 \\
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For the 10 numbers: 43, 35, 43, 33, 38, 53, 64, 27, 34, 27

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- $\text{IQR} = Q_3 - Q_1 = 43 - 33 = 10$
- Five number summary: 27, 33, 36.5, 43, 64

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For the 10 numbers: 43, 35, 43, 33, 38, 53, 64, 27, 34, 27

$$
\begin{array}{ccccc}
43 & & 27 & & \\
35 & & 27 & & \\
43 & & 33 & & \\
33 & & 34 & & \\
38 & \xrightarrow{\text{sort}} & 35 & & \\
53 & & 38 & \leftarrow & \text{overall median} = \dfrac{35+38}{2} = 36.5 = Q_2 \\
64 & & 43 & & \\
27 & & 43 & & \\
34 & & 53 & & \\
27 & & 64 & &
\end{array}
$$

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64		43	
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53		38	
64		43	
27		43	$\leftarrow$ median of this half $= 43 = Q_3$
34		53	
27		64	

- IQR $= Q_3 - Q_1 = 43 - 33 = 10$
- Five number summary: 27, 33, 36.5, 43, 64

Calculation of Quartiles

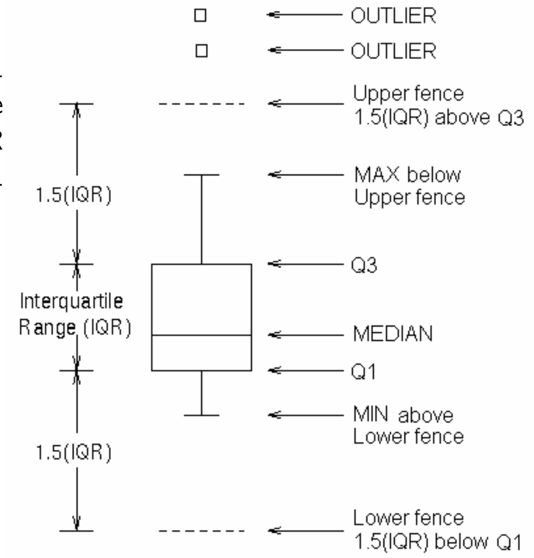
Don't worry about the calculation of quartiles. Just keep in mind that

$\boxed{\text{quartiles divide data into 4 even parts}}$

Box-and-Whiskers Plot (also called Boxplot)

1.5 IQR Rule

Recall IQR = Q3 − Q1 observations that are more than 1.5 IQR above Q3 or below Q1 are (potential) outleirs



Boxplot v.s. Histogram — Skewness

What does a boxplot look like if the histogram is symmetric?

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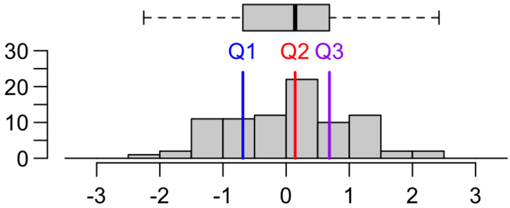

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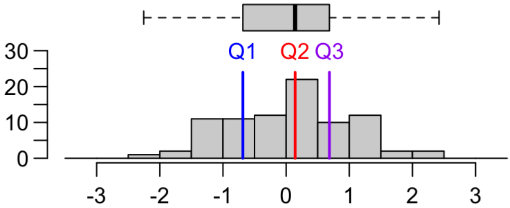

Ditto, if right-skewed?

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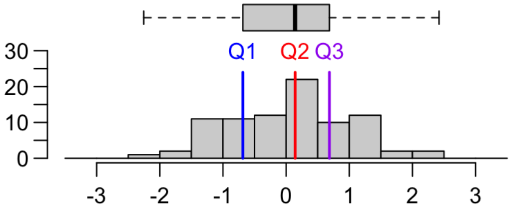

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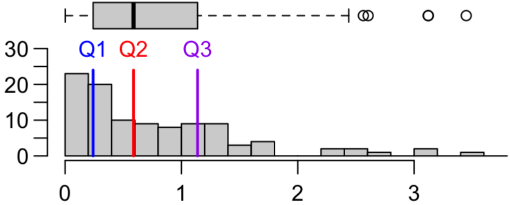

Boxplot v.s. Histogram — Modality

Can you tell from a boxplot whether the distribution is unimodal or bimodal?

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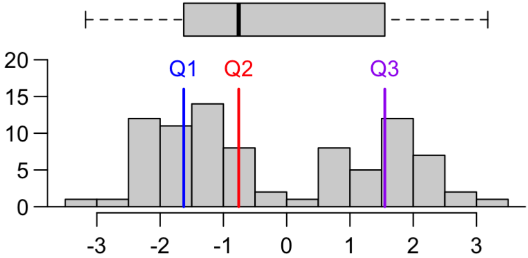


Back to FEV Data

Example: FEV Data

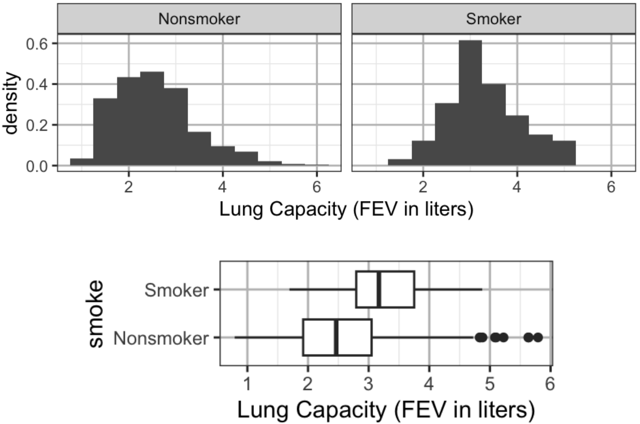
The FEV data are data of a sample of 654 youths, aged 3 to 19, in the area of East Boston during middle to late 1970's. The variables include

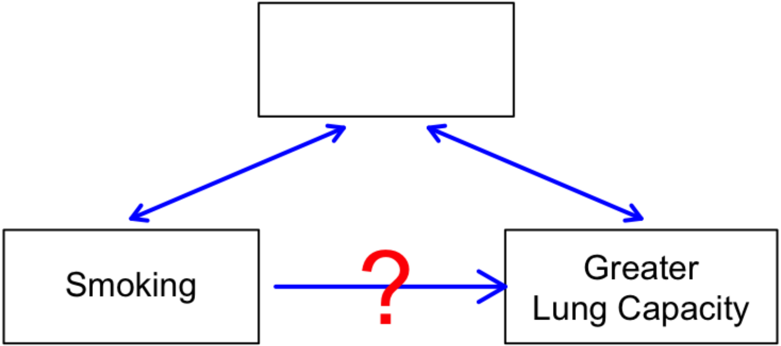
- `age`: subject's age in years
- `fev`: lung capacity of subject, measured by **forced expiratory volume** (abbreviated as **FEV**), the amount of air an individual can exhale in the first second of forceful breath in liters
- `ht`: subject's height in inches
- `sex`: gender (0 = Female, 1 = Male)
- `smoke`: smoking status (0 = Nonsmoker, 1 = Smoker)

```
  age   fev   ht sex smoke
    9 1.708 57.0   0     0
    8 1.724 67.5   0     0
    7 1.720 54.5   0     0
...(omitted)...
   15 3.211 66.5   0     0
```

Smokers had greater lung capacities than nonsmokers on average in the FEV study. *Smoking increases one's lung capacity!?!*

smoke	min	Q1	median	Q3	max	mean	sd	n
Nonsmoker	0.791	1.920	2.465	3.048	5.793	2.566143	0.8505215	589
Smoker	1.694	2.795	3.169	3.751	4.872	3.276862	0.7499863	65



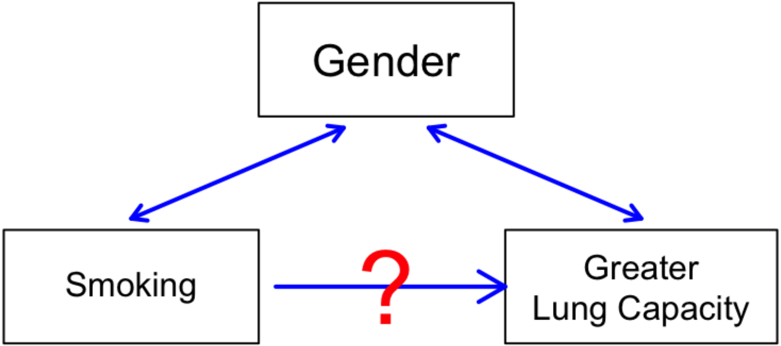

Smoking
?
Greater
Lung Capacity

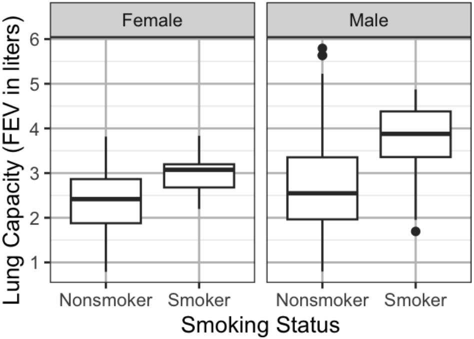
Controlling for Confounders in Observational Studies

- Recall we can never make a *cause-and-effect* conclusion from an observational studies due to confounding
- However, just because confounding is possible in observational studies does not mean that investigators are powerless to address it
- Instead, well-conducted observational studies make strong efforts to identify confounders and *control for their effects*
- There are many techniques for doing so; the most direct approach is to *make comparisons separately for smaller and more homogeneous groups*

- For example, studying the association between heart disease and smoking could be misleading, because men are more likely to have heart disease and also more likely to smoke
- A solution is to compare heart disease rates separately for the two genders
    - male smokers v.s. male nonsmokers, and
    - female smokers v.s. female nonsmokers
- Age is another common confounding factor that epidemiologists are often concerned with controlling for

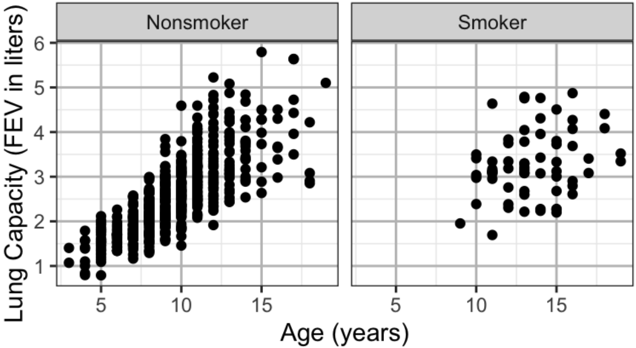
Is Gender a Confounder of the FEV Study?





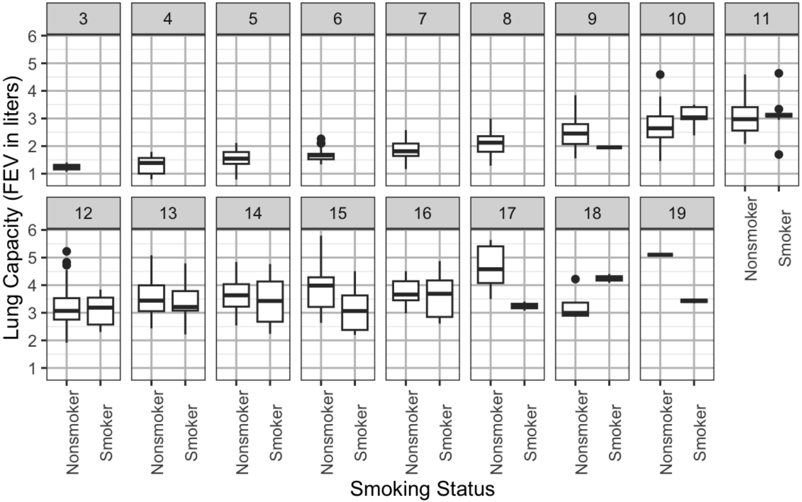
For females, did smokers or nonsmokers have greater lung capacity? How about males?

Age Confounded Smoking Effect on Lung Capacities



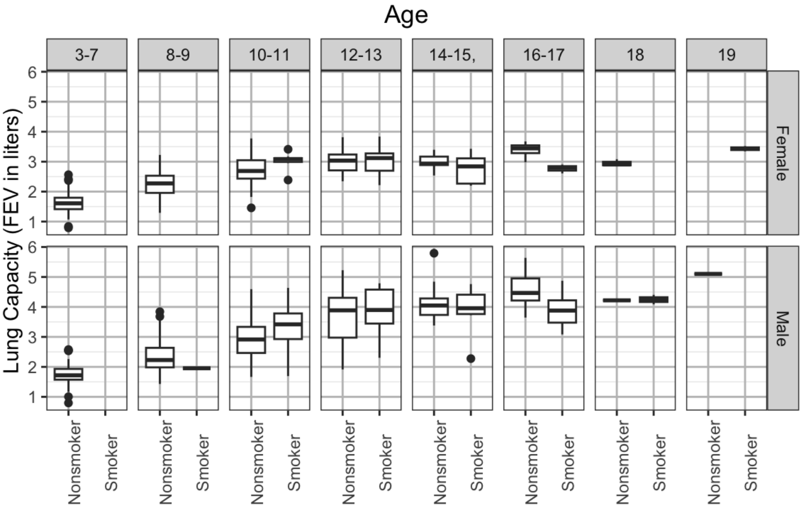
- Older children had greater lung capacities
- Smokers are older (mostly teens) while nonsmokers include children and teens

Effect of Smoking on FEV After Accounting for Age



Considering children of the same age, did smokers or nonsmokers had greater lung capacities in general?

Effect of Smoking on FEV After Accounting for Age & Sex



Value of Observational Studies

- It's possible to reveal the true relations between variables in observational studies, if we inspect the data carefully, controlling for potential confounding variables (making comparisons separately for each level of the confounding variable).

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- Observational studies have tremendous value as initial studies to build up support for larger, more resource-intensive controlled experiments
- However, they could be *misleading* – identifying confounders is not always easy, and you can't control for everything
- Media often misinterpret association found in an scientific study as causation. Be cautious when reading news reports.

Simpson's Paradox

Simpson's Paradox is the phenomenon that a trend observed in separate groups of data might disappear or reverse when the groups are combined.

Example: The FEV study is an example of Simpson's Paradox

- For each age group, nonsmokers had greater lung capacities in general than smokers
- Combining all age groups, smokers had greater lung capacities in general than nonsmokers

Example: Smoking and Longevity

A survey during 1972-74 recruited 1314 women in the United Kingdom and asked if they smoked. A follow-up survey 20 years later determined whether each woman was deceased or still alive.

	Dead	Alive	Percentage Died
Smoker	139	443	$139/(139+443) \approx 23.9\%$
Nonsmoker	230	502	$230/(230+502) \approx 31.4\%$

Surprisingly, smokers had a lower death rate than nonsmokers

Example: Smoking and Longevity, Controlling for Age

Age in 1972	Smoke?	Dead	Alive	Total	% Dead
18-34	Y	5	174	179	$5/179 \approx 2.8\%$
	N	6	213	219	$6/219 \approx 2.7\%$
35-54	Y	41	198	239	$41/239 \approx 17.2\%$
	N	19	180	199	$19/199 \approx 9.5\%$
55-64	Y	51	64	115	$51/115 \approx 44.3\%$
	N	40	81	121	$40/121 \approx 33.1\%$
65+	Y	42	7	49	$42/49 \approx 85.7\%$
	N	165	28	193	$165/193 \approx 85.5\%$

With each age group, smokers had a higher death rate than nonsmokers.

How could smokers have a lower death rate than nonsmokers when all age groups are combined?

Simpson's Paradox

Age in 1972	Smoke?	Dead	Alive	Total	% Dead	% Smoke
18-34	Y	5	174	179	2.8%	$\frac{179}{179+219} \approx 45.0\%$
	N	6	213	219	2.7%	
38-54	Y	41	198	239	17.2%	$\frac{239}{239+199} \approx 54.6\%$
	N	19	180	199	9.5%	
55-64	Y	51	64	115	44.3%	$\frac{115}{115+121} \approx 48.7\%$
	N	40	81	121	33.1%	
65+	Y	42	7	49	85.7%	$\frac{49}{49+193} \approx 20.2\%$
	N	165	28	193	85.5%	

- Old women (65+) were mostly dead 20 years later.

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- Old women (65+) were mostly dead 20 years later.
- There were fewer smokers among old women

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- Old women (65+) were mostly dead 20 years later.
- There were fewer smokers among old women
- Smokers had a lower death rate because they were generally younger — age is a confounder here.

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- Old women (65+) were mostly dead 20 years later.
- There were fewer smokers among old women
- Smokers had a lower death rate because they were generally younger — age is a confounder here.
- *Simpson's paradox*: a trend observed in different groups of data can disappear or reverse when the groups are combined.

Sex Bias in Graduate Admissions

UC Berkeley admission rate:

	Admitted	Rejected	Percentage Admitted
Male	1198	1493	$1198/(1198 + 1493) \approx 44.5\%$
Female	557	1278	$557/(557 + 1278) \approx 30.4\%$

- Sex bias?
- The data is observational — caution is in order.
- This comparison is made over the whole university. Why not look at each individual department, which is smaller and more homogeneous?

Sex Bias in Graduate Admissions (Cont'd)

	Men			Women		
Dept	Number Admitted	Number Rejected	Percent Admitted	Number Admitted	Number Rejected	Percent Admitted
A	512	313	62%	89	19	82%
B	353	207	63%	17	8	68%
C	120	205	37%	202	391	34%
D	138	279	33%	131	244	35%
E	53	138	28%	94	299	24%
F	22	351	6%	24	317	7%
Total	1198	1493	44.5%	557	1278	30.4%

Simpson's paradox:

- In major A, admission rate: **female** $>$ **male**
- In other majors, admission rate: **female** $\approx$ **male**
- Overall admission rate: **female** $<$ **male**
- Why is that?

Sex Bias in Graduate Admissions (Cont'd)

- More men apply for "easy majors" (A and B),
  which raise the total number of admitted men, even though the admission rates for men are lower than women;
- Most women applied for "hard majors" (C, D, E, F),
  which lowers the total number of admitted women, even though the admission rates for women are roughly equal to men.
- Confounding factor — majors
- Within-major comparisons lead to the correct conclusion. *A way to rule out the effect of a confounder is to **control for** it. That is, break the subjects by the confounder into groups, and make separate comparisons in all groups.